EXPERIMENTAL AND ARTIFACT PLAN

CERA

Cross-Execution Recovery through Effect Adoption for Stochastic Agents

SOSP 2027

Purpose. This document specifies the executable evaluation contract for CERA-Bench generation, E0A–E0B, and E1–E5: workloads, models, seeds, resource semantics, independent oracles, algorithms under test, inputs and outputs, formulas, statistical procedures, acceptance criteria, and artifact flow.

Evaluation principle. LLMs generate stochastic executions; they never define semantic truth. CERA-Bench is frozen before system comparison. E1 uses an independent oracle. E2 and E3 consume identical frozen E0A traces. E0B is external realism evidence only. Unsupported semantic evidence is represented as UNKNOWN, never as BLOCK by default.

1 Evaluation Scope and Formal Contract

The evaluation targets the post-commit escape regime. An original execution X_o externalizes effects before a fault; recovery produces X_n, potentially with a different causal graph. For each historical effect $e \in H_o$, the semantic oracle determines

$$\text{Compat}(e, X_o, X_n) \in \{\text{true}, \text{false}, \text{undetermined}\},$$

corresponding to ADOPT, BLOCK, and UNKNOWN.

$$\text{Compat}(e, X_o, X_n) \iff \forall X \in C(X_n, R), \quad \text{Post}(e) \text{ remains true under } X, \tag{1}$$

where $C(X_n, R)$ is the set of admissible continuations permitted by the supported resource semantics and the authority/anchor obligations of e.

For a divergent event set Δ, define the conservative influence cone

$$\text{Infl}(\Delta) = \text{OverApprox}_R\big(\text{Reach}(\Delta)\big), \tag{2}$$

where the resource adapter provides the sound projection from causal provenance to resource influence.

The trusted semantic components are the resource adapters, independent oracle, compatibility engine, exact history selector for the supported conflict class, and future-effect admission boundary. LLM outputs, tool arguments, natural-language intent, opaque resources, and unauthoritative trace text are untrusted.

2 CERA-Bench and Workload Design

2.1 Benchmark object

Each benchmark case is

$$B = \langle X_o, f, X_n, E_o, E_n, GT, \Pi \rangle, \tag{3}$$

where X_o is the original execution, f is the declared fault/divergence condition, X_n is the recovery execution, E_o, E_n are historical/new effects, GT is the independent oracle, and Π records provenance and authoritative resource evidence.

The oracle is defined from resource transitions, identity, version/epoch anchors, invariants, and atomic validation semantics. It is implemented independently of the CERA checker and never calls an LLM judge.

2.2 Workload families

Family	Canonical workloads	Primary resources
Software engineering	build/release, dependency-update/build, patch/test/publish	Filesystem, PostgreSQL, object store
Deployment/state transition	artifact promotion, staged release, configuration rollout	Object store, PostgreSQL, transactional service
Data workflow	ETL publication, incremental pipeline, validation/publication, versioned record update	PostgreSQL, object store
Transactional workflow	reservation/approval, staged update, multi-step state transition	PostgreSQL, transactional service

Twelve canonical templates are instantiated into 120 task specifications, 30 per family.

2.3 Supported resource semantics

Filesystem. Identity is a path/object identity plus workspace scope; content state is anchored by revision or digest; atomic comparison/update is required whenever concurrent mutation can invalidate evidence.

PostgreSQL. Identity is table/row plus transaction context; state is anchored by transaction/version metadata; serializable or CAS-like revalidation is used where required by the effect class.

Object store. Identity is object plus bucket/prefix scope; immutable object versions/digests and conditional writes provide authoritative anchors.

Transactional service. Identity is reservation/approval/workflow identity; authority and workflow version are explicit; supported stage transitions are monotone where specified.

3 Models, Sampling, and Benchmark Generation

3.1 Models

The workload-generation set contains four open-weight local models.

Model	Role	Use
Qwen3-8B	Primary	Full benchmark generation; primary E0A results.
Qwen3-14B	Capacity robustness	Full B0 generation; E0A robustness.
Qwen3-32B	Higher-capacity robustness	Full B0 generation; E0A robustness.
Llama-3.1-8B-Instruct	Cross-family robustness	Full B0 generation; E0A robustness.

Exact model revisions, tokenizer revisions, inference framework, quantization, decoding parameters, and license metadata are pinned in the experiment manifest. Model weights are not redistributed as benchmark artifacts.

3.2 Sampling budget

B0 generates

$$120 \text{ tasks} \times 4 \text{ models} \times 3 \text{ seeds} = 1,440 \tag{4}$$

frozen stochastic cases. Seeds are $\{42, 43, 44\}$. A 30-task task-level evaluation slice is reserved for the primary E0A/E2/E3 corpus; task identity, not trace identity, defines evaluation separation.

E0A uses the same 30 tasks, two fixed seeds $\{42, 43\}$, four models, and one fault per case:

$$30 \times 4 \times 2 = 240 \text{ primary controlled cases.} \tag{5}$$

The third B0 seed remains available for artifact-level robustness analysis and is not introduced into the primary E0A/E2/E3 comparison unless explicitly declared before analysis.

3.3 Benchmark-generation algorithm

Algorithm 1 CERA-Bench generation

Require: Task specification T, model M, seed s, resource adapters R, fault policy F
Ensure: Frozen benchmark case B

1: Instantiate T in a clean local environment.
2: Execute the original agent trajectory and record managed effects and provenance.
3: Require at least two committed external effects before the candidate fault point.
4: Select the fault independently of any future recovery output and inject $f \in F$ after effects have escaped.
5: Construct the declared recovery state and sample a recovery trajectory with (M, s).
6: Record all historical/new effects, anchors, authority, and resource state.
7: Compute GT from resource semantics, never from CERA or an LLM judge.
8: Accept B only if trace, fault, effect, oracle, and reproducibility checks pass.

3.4 Benchmark integrity gates

A case is accepted only if (i) event structure and causal parents are valid; (ii) step/timestamp order is monotone; (iii) the fault occurs after at least two committed effects; (iv) every managed effect has identity, anchor, and postcondition information; (v) the oracle package is independent of CERA; (vi) task-level separation is preserved; and (vii) the original case is reproducible within a declared tolerance under the pinned environment hash, model revision, and seed.

4 E0A: Controlled Post-Commit Recovery

4.1 Question and purpose

E0A asks whether stochastic recovery after escaped effects produces causal divergence for which some historical effects remain valid while others do not. The primary quantities are

$$P(G_n \neq G_o), \qquad P(\text{mixed compatibility} \mid G_n \neq G_o). \tag{6}$$

The experiment is not a population estimate over all agents; it is a controlled stress study with exact semantics.

4.2 Inputs and outputs

Input	Source
30 task specifications	`benchmark/tasks/`
Frozen original/recovery traces and state snapshots	`benchmark/traces/`
Independent compatibility labels	`benchmark/oracles/`
Model/seed/fault manifest	`benchmark/manifests/`

The notebook emits machine-readable per-case records and aggregate summaries to `results/E0A/` and publication figures to `figures/E0A/`.

4.3 Primary calculations

Let A_o, A_n be the action sequences of the original and recovered executions. Define

$$D_{\text{action}} = 1 - \frac{|\text{LCS}(A_o, A_n)|}{\max(|A_o|, |A_n|)}. \tag{7}$$

Let Δ be the unmatched/ambiguous event set. Define

$$\kappa = \frac{|\text{Cone}(\Delta)|}{|V_o| + |V_n|}, \qquad \eta = \frac{|H_o|}{N_{\text{effects before failure}}}. \tag{8}$$

For a case,

$$\text{MixedCompatibility} = \mathbf{1}\left[\exists e_i : \text{state}(e_i) = \textsf{ADOPT} \wedge \exists e_j : \text{state}(e_j) = \textsf{BLOCK}\right], \tag{9}$$

and

$$\text{AdoptionOpportunity} = \frac{\#\textsf{ADOPT}}{\#\textsf{ADOPT} + \#\textsf{BLOCK}}, \tag{10}$$

with UNKNOWN excluded from the denominator.

For a retained set $A \subseteq H_o$,

$$\text{ValidWorkRetained} = \sum_{e \in A, GT(e)=\textsf{ADOPT}} V(e), \tag{11}$$

$$\text{InvalidRetention} = |\{e \in A : GT(e) = \textsf{BLOCK}\}|, \tag{12}$$

$$\text{UnnecessaryDiscard} = \sum_{e \notin A, GT(e)=\textsf{ADOPT}} V(e). \tag{13}$$

4.4 Sampling and reporting

The primary controlled suite contains 240 cases: 30 tasks, four models, two seeds. Report per-model and per-family rates, Wilson 95% confidence intervals for binary outcomes, and mean, median, IQR, minimum, and maximum for continuous divergence measures. Model comparisons are paired by task ID and seed; shared tasks/environments are not treated as independent samples.

5 E0B: External Divergence Study

5.1 Question and dataset

E0B asks whether repeated real agent executions of the same external task materially diverge. The dataset is the OSWorld-V2 repeated-run corpus. The primary study uses 30 trace-quality-qualified same-task pairs, selecting each task once; the two directions of the same pair are not counted as independent observations.

5.2 Inputs and outputs

Input	Output
OSWorld run1/run2 trajectories from `data/osworld/`	Pair-level action divergence.
Trace metadata	Observable external-effect disagreement where evidence is sufficient.
No CERA semantic oracle	No ADOPT/BLOCK claim is inferred from OSWorld trace text.

$$\text{MaterialDivergence} = \mathbf{1}[D_{\text{action}} \geq 0.25]. \tag{14}$$

Report the task-level rate with a Wilson 95% interval and the distribution of D_{action}. Missing effect evidence remains UNKNOWN rather than BLOCK.

6 E1: Semantic Correctness and Exact Selection

6.1 Question

E1 verifies that the compatibility engine matches an independently computed oracle and that the exact history selector returns the optimal safe subset for the supported compositional conflict class.

6.2 Deterministic semantic suite

Construct 640 deterministic cases, 80 per class:

Class	Cases	Required semantics
P1 Non-interference	80	Disjoint footprint and valid anchors.
P2 Restricted identity preservation	80	Stable identity/authority with invariant obligation.
P3 Monotone preservation	80	Upward-closed obligation under $\preceq_R$.
P4 Atomic revalidation	80	CAS/version/epoch/lease or equivalent atomic validation.
Stale/invalid evidence	80	Snapshot-only or invalidated evidence must not yield ADOPT.
Cross-effect conflicts	80	Safe subset selection under explicit conflicts.
New-effect admission	80	Valid effects admitted; invalid/unsupported effects blocked.
UNKNOWN/unsupported	80	Insufficient evidence maps to UNKNOWN.

For every case, CERA's output is compared with the independent oracle.

6.3 Exact-selection suite

Generate 200 bounded conflict instances with at most 10 historical effects. Enumerate all $2^{|H_o|}$ subsets, define utility $V(e)$ independently, and compute

$$A^* = \arg\max_{A \subseteq H_o} \sum_{e \in A} V(e) \quad \text{s.t.} \quad \text{Safe}(A, X_o, X_n). \tag{15}$$

Compare against CERA's selected set A_{CERA}:

$$\text{SelectionAgreement} = \mathbf{1}[A_{\text{CERA}} = A^*], \tag{16}$$

$$\text{UtilityRegret} = \sum_{e \in A^*} V(e) - \sum_{e \in A_{\text{CERA}}} V(e), \tag{17}$$

$$\text{InvalidRetention} = |\{e \in A_{\text{CERA}} : GT(e) = \text{BLOCK}\}|. \tag{18}$$

6.4 Acceptance criteria

The supported semantic suite must have zero invalid retention; the exact-selector suite must have 100% selection agreement and zero utility regret; UNKNOWN must appear only where the independent oracle deems evidence insufficient or unsupported. All formal counterexamples are regression tests.

7 E2: Main Recovery Comparison

7.1 Question

Does effect-level adoption retain more independently valid historical work than strong recovery baselines without unsafe retention?

7.2 Frozen inputs

The comparison uses the 240 frozen E0A cases. No baseline regenerates agent trajectories. Every method receives the same original/recovery traces, fault, effect history, and observation budget; observation differences are varied only in E3.

7.3 Systems

System	Policy
Replay/rollback	Re-execute the old trajectory or discard history at the recovery boundary according to the baseline semantics.
DART-style	Select a legal recovery boundary without explicit per-effect cross-execution compatibility.
DART+	Strong effect-retention implementation under matched metadata, without the CERA preservation calculus.
Selective recovery	Dependency/taint-guided retention/re-execution baseline.
CERA	Cross-execution compatibility, exact supported history selection, and admission of new managed effects.

7.4 Metrics

$$\text{ValidWorkRetained} = \sum_{e \in A, GT(e)=\text{ADOPT}} V(e), \tag{19}$$

$$\text{InvalidRetention} = |\{e \in A : GT(e) = \text{BLOCK}\}|, \tag{20}$$

$$\text{UnnecessaryDiscard} = \sum_{e \notin A, GT(e)=\text{ADOPT}} V(e), \tag{21}$$

$$\text{RecoveryCost} = C_{\text{reexec}} + C_{\text{cert}} + C_{\text{prov}}, \tag{22}$$

$$\text{RecoveryEfficiency} = \frac{\text{ValidWorkRetained}}{\text{RecoveryCost}}. \tag{23}$$

The primary plot is the three-way frontier with invalid retention on the horizontal axis, valid work retained on the vertical axis, and recovery cost encoded by point size. The positive result is a non-empty regime in which CERA lies on the efficient frontier with zero invalid retention on supported cases.

8 E3: Provenance Gain

8.1 Question

Does authoritative resource/OS evidence materially change the safety decision for a historical effect?

8.2 Observation modes

Mode	Visible evidence
$O_{\text{framework}}$	Agent/framework events only.
O_{tool}	Framework plus tool/runtime events.
O_{resource}	Tool evidence plus resource identity/version/state.
O_{OS}	Combined framework, resource, and selected OS/process provenance.

Run the same 240 E0A cases under masked observation views; do not regenerate trajectories.

8.3 Metrics

$$\text{ObservationGain} = \#\{\text{correct decision changes induced by added evidence}\}, \tag{24}$$

$$\text{SafetyGain} = \#(\text{ADOPT}_{\text{framework}} \rightarrow \{\text{BLOCK}, \text{UNKNOWN}\}_{\text{OS}} \text{ where } GT = \text{BLOCK}), \tag{25}$$

$$\text{UsefulGain} = \#(\text{UNKNOWN}_{\text{framework}} \rightarrow \text{ADOPT}_{\text{OS}} \text{ where } GT = \text{ADOPT}), \tag{26}$$

$$\text{FalseConfidence} = \#(\text{ADOPT}_{\text{framework}} \text{ with } GT = \text{BLOCK} \text{ before added evidence}). \tag{27}$$

At least five representative cases must expose a machine-readable evidence chain from framework event to tool event to resource state to OS/process evidence, together with the resulting decision transition and oracle outcome.

9 E4: Operating Regime

9.1 Question

When does CERA provide a positive preservation/cost trade-off, and when should it conservatively give up?

9.2 Parameter grid

Parameter	Levels	Interpretation
Externalization ratio η	0.10, 0.30, 0.50, 0.80	Fraction of work externalized before failure.
Divergence density κ	0.10, 0.30, 0.50, 0.80	Fraction of execution influenced by divergence.
Resource opacity	low, medium, high	Fraction of relevant state not directly observable by the adapter.
Conflict density	low, high	Density of cross-effect interactions.
Cases/configuration	10	Independent deterministic variants per configuration.

This yields

$$4 \times 4 \times 3 \times 2 = 96 \text{ configurations}, \qquad 96 \times 10 = 960 \text{ evaluated cases}. \tag{28}$$

For one-factor plots, vary one parameter while holding the others at their nominal midpoint; use the full factorial for the final heatmap/frontier. Deterministic seeds are fixed within each configuration.

9.3 Expected qualitative regime

At low escape, containment/replay may dominate. Under moderate escape and localized divergence, CERA should preserve more valid work. As κ approaches dense divergence, influence cones expand and CERA should become more conservative. Increasing resource opacity should increase UNKNOWN/ESCALATE rather than unsafe ADOPT.

10 E5: Cost and Adversarial Safety

10.1 Safety suite

Use 20 cases for each of the following ten classes: TOCTOU/version churn, concurrent resource modification, hidden subprocess access, inherited descriptors, unmanaged sockets/effects, capability revocation, delayed validation, duplicate recovered actions, provenance reordering, and admission-boundary bypass. The safety suite therefore contains 200 cases.

10.2 Performance suite

Measure 50 representative workloads with five deterministic repetitions each, yielding 250 runtime runs. Report warm-cache and cold-cache conditions for runtime measurements and keep semantic outcomes fixed across repetitions.

10.3 Metrics

$$\text{ProvenanceCPU} = \frac{T_{\text{CERA,CPU}} - T_{\text{baseline,CPU}}}{T_{\text{baseline,CPU}}}, \tag{29}$$

$$\text{MetadataOverhead} = \frac{B_{\text{provenance}}}{N_{\text{events/effects}}}, \tag{30}$$

$$\text{AlignmentLatency} = T_{\text{align}} - T_{\text{recovery start}}, \tag{31}$$

$$\text{CertificationLatency} = T_{\text{decision}} - T_{\text{alignment complete}}, \tag{32}$$

$$\text{AdmissionLatency} = T_{\text{admit}} - T_{\text{effect ready}}, \tag{33}$$

$$\text{EndToEndRecovery} = T_{\text{isolation}} + T_{\text{align}} + T_{\text{cert}} + T_{\text{execution}} + T_{\text{admission}}. \tag{34}$$

10.4 Safety criteria

The supported adversarial suite must have zero unsafe ADOPT. Unsupported or stale evidence must become UNKNOWN, BLOCK, or ESCALATE; no unsupported effect may escape the admission boundary before validation. Version churn invalidates stale evidence unless atomic revalidation or a monotonicity proof establishes safety.

11 Repository and Experiment Data Flow

The repository is intentionally small:

```
cera/
  README.md
  requirements.txt
  LICENSE
  configs/
    benchmark.yaml
    models.yaml
    experiments.yaml
```

```
cera/
    README.md
    requirements.txt
    LICENSE
    configs/
        benchmark.yaml   models.yaml   experiments.yaml
    notebooks/
        generate_cera_bench.ipynb
        E0A_controlled_recovery.ipynb
        E0B_external_divergence.ipynb
        E1_semantic_correctness.ipynb
        E2_recovery_comparison.ipynb
        E3_provenance_gain.ipynb
        E4_operating_regime.ipynb
        E5_cost_and_safety.ipynb
    benchmark/
        tasks/   traces/   oracles/   manifests/
    data/
        osworld/
    results/
        E0A/  E0B/  E1/  E2/  E3/  E4/  E5/
    figures/
        E0A/  E0B/  E1/  E2/  E3/  E4/  E5/
```

11.1 Producer-consumer contract

Stage	Reads	Writes	Consumed by
B0	task templates, resource adapters, model configs, seeds	`benchmark/tasks/`, `benchmark/traces/`, `benchmark/oracles/`, `benchmark/manifests/`	E0A, E1, E2, E3, E4, E5
E0A	frozen benchmark subset + oracle	`results/E0A/`, `figures/E0A/`	E2, E3
E0B	`data/osworld/`	`results/E0B/`, `figures/E0B/`	Paper only
E1	deterministic semantic cases + independent oracle	`results/E1/`, `figures/E1/`	Paper only
E2	frozen E0A corpus + baseline implementations	`results/E2/`, `figures/E2/`	Paper
E3	frozen E0A corpus + observation masks	`results/E3/`, `figures/E3/`	Paper
E4	task semantics + regime generator	`results/E4/`, `figures/E4/`	Paper
E5	representative benchmark cases + fault schedules	`results/E5/`, `figures/E5/`	Paper

The only experiment-to-experiment data dependencies are $E0A \rightarrow E2$ and $E0A \rightarrow E3$. E1 is independent correctness validation. E4 generates controlled regime variants from frozen task semantics rather than regenerating model trajectories. E5 reuses representative workloads but changes execution schedules and faults. No notebook overwrites `benchmark/`; each experiment writes only to its own result/figure directory.

12 Theory-to-Code Contract

The implementation and evaluation must preserve the following semantic rules:

$$P1: \quad F_e \cap \mathrm{Infl}(\Delta) = \varnothing \ \wedge \ \mathrm{AnchorsValid}(e) \Longrightarrow \mathsf{ADOPT}, \tag{35}$$

$$P2: \quad \mathrm{SameId}(e) \wedge \mathrm{AuthorityValid}(e) \wedge \mathrm{EffectInvariant}(e) \Longrightarrow \mathsf{ADOPT}, \tag{36}$$

$$P3: \quad s_o \preceq_R s_n \wedge \mathrm{UpwardClosed}(\mathrm{Post}(e)) \Longrightarrow \mathsf{ADOPT}, \tag{37}$$

$$P4: \quad \mathrm{AtomicRevalidation}(e) \Longrightarrow \mathsf{ADOPT}. \tag{38}$$

If incompatibility is established, return BLOCK; otherwise return UNKNOWN. A stale snapshot alone never produces ADOPT.

Algorithm	Evaluation contract
Conservative cross-execution alignment	Match original and recovered events by stable execution sites and resource anchors; reject incompatible parents/identities/versions; treat unmatched or ambiguous events as divergence.
Influence analysis	Compute a sound over-approximation Infl(Δ) of resources influenced by divergent events using the resource adapters.
Historical-effect adoption	For each $e \in H_o$, derive P1–P4; output ADOPT when a witness exists, BLOCK when incompatibility is proved, and UNKNOWN otherwise.
Safe-history selection	Construct the supported cross-effect conflict graph, solve each supported independent conflict group exactly, and escalate unsupported interaction structures.
Future-effect admission	Keep recovered effects isolated until capability, anchor, retained-history compatibility, and atomicity checks succeed; suppress failures.

12.1 CERA execution pipeline

Algorithm 2 CERA recovery evaluation path

Require: Frozen original execution X_o, fault f, recovered execution X_n, evidence Π
1: Isolate the recovered execution and load immutable historical ledger H_o.
2: Align G_o and G_n conservatively and compute Δ.
3: Compute Infl(Δ) using the resource adapters.
4: For each $e \in H_o$, derive P1–P4 and assign ADOPT, BLOCK, or UNKNOWN.
5: Solve supported cross-effect conflict groups to obtain A^*; escalate unsupported structures.
6: Admit each new managed effect only after capability, anchor, compatibility, and atomicity checks.
7: Record all decisions and evidence chains.

13 Statistical Analysis and Reporting

Binary rates use Wilson 95% confidence intervals. Paired model/system comparisons use paired bootstrap intervals or an exact paired test over shared task IDs; report effect size and confidence interval. Continuous latency/overhead distributions are summarized with median, IQR, and p95; mean-only summaries are avoided for heavy-tailed measures. Model robustness is first reported per workload family and then pooled. No safety metric uses an LLM judge.

For E0B, $n = 30$ is the number of independent same-task pairs. For E0A, $n = 240$ is the number of model-task-seed cases, with model and task repeated-measures structure reported explicitly. Shared environments or seeds are never counted as independent replications merely by producing multiple traces.

14 Required Figures, Tables, and Machine-Readable Outputs

ID	Content	Source
Fig. 1	CERA execution model: old history, fault, divergent recovery, adoption	Theory
Fig. 2	E0A divergence and mixed-compatibility distribution	E0A
Fig. 3	Valid work retained versus invalid retention for CERA and baselines	E2
Fig. 4	Recovery-efficiency frontier	E2/E4
Fig. 5	Verified observation-gain transitions	E3
Fig. 6	η-κ operating-regime heatmap	E4
Fig. 7	Runtime overhead breakdown	E5
Fig. 8	Adversarial safety outcomes	E5
Table 1	Workload/resource/model matrix	B0
Table 2	Semantic correctness and exact selection	E1
Table 3	Main recovery comparison	E2
Table 4	Model robustness	E0A
Table 5	Observation-gain cases	E3

Every experiment writes a machine-readable manifest containing the benchmark hash, code commit, model revision, quantization, seed, environment hash, adapter version, and execution timestamp. Aggregate figures are generated only from these result manifests.

15 Execution Order and Acceptance Gates

Stage	Action	Exit criterion
B0	Freeze task/resource/model definitions and generate CERA-Bench	1,440 valid cases with independent oracles.
E0A	Run the 240 controlled cases	Divergence characterized and mixed-compatibility regime either demonstrated or subjected to the explicit kill/narrow decision.
E0B	Run 30 OSWorld pairs	Divergence statistics complete; no semantic overclaim.
E1	Run correctness and exact-selection suites	Zero invalid retention; 100% exact-selection agreement on supported class.
E2	Run all baselines on identical frozen E0A cases	Positive preservation/cost regime with zero invalid retention for supported cases.
E3	Compare observation masks	At least one independently verified decision gain from authoritative evidence.
E4	Sweep operating parameters	Useful regime and conservative failure envelope identified.
E5	Run safety/performance suites	Zero unsafe ADOPT; no managed new-effect leak; overhead quantified.

The no-regeneration rule is strict: after B0 is frozen, E1–E5 do not regenerate stochastic traces except the explicitly controlled parameterized variants of E4. E2 and E3 consume exactly the frozen 240-case E0A corpus. Any deviation from the frozen manifest requires a new manifest version and must not silently replace the primary result.

Submission criterion. The experimental program supports the CERA thesis only if the independent oracle confirms zero invalid adoption on supported cases, the main comparison demonstrates a positive valid-work-preservation/cost trade-off, E3 demonstrates at least one correctness-changing observation gain, E4 exposes a reproducible operating envelope, and E5 establishes fail-closed behavior under adversarial conditions.